VR-Forms — v1.0.3 (revision note)

Two editorial corrections (VR v1.1.1 sync; "no ontology" scoped) and one added section: Things and Acts

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Scope. This is the revision delta for VR-Forms v1.0.3. *No construction, definition, theorem, or axiom-profile of VR-Forms is altered.* Two editorial corrections and one added expository section: (1) the recall of the parent work VR is brought up to date with VR v1.1.1; (2) the "no ontology" language is scoped to "no ontology of substance"; (3) a new section, *Things and Acts*, is added to the two-register discussion — it names, and prices, what the two registers already hold, resting on the cycle's existing axiom profiles (no new formal content).

(1) Correction: the parent-VR recall

The introduction described VR as

> "a minimal axiomatisation of arithmetic on **three primitives** $\{\emptyset, \to, t\}$ and **four axioms**; equivalent to Peano arithmetic."

This is superseded. In VR v1.1.1 the propositional implication $\to$ (and the principles A1, A2 and the two-valued logical register) are **removed**: $\to$ generated only a finite truth-function algebra used by no VR theorem — arithmetic, not logic — and VR generates no logic of its own (the logic it reasons with is metatheoretic, borrowed). The corrected recall reads

> "a minimal *axiom-free* reconstruction of arithmetic on **two primitives** $\{\emptyset, t\}$ — the nullary base $\emptyset$ and the unary succession t — with no axioms of its own (membership is defined; the succession principle A3 is a theorem; induction A4 is the recursor); equivalent to first-order Peano arithmetic (Zenodo DOI 10.5281/zenodo.20688164, v1.1.1; concept 10.5281/zenodo.20212091)."

(2) Correction: scoping the "no ontology" language

Several places in VR-Forms state, without qualification, that "the cycle ascribes no ontology" (and the "ontological register" label is dropped on that basis). Read at face value this is a *blanket* denial of ontology, and it collides with the global membership relation $\in$ that VR-Sets does carry. The intended and accurate claim is narrower: VR ascribes no ontology *of substance* — no object that *is* prior to operations; being is relocated into doing. It is not the claim that the cycle ascribes no ontology *at all*: the operational/set layer carries genuine relational and structural commitments (global $\in$, potentialist infinity, relational identity), which are *tracked* (the two registers and the axiom tiers), not denied. Accordingly every unqualified "ascribes no ontology" should read "ascribes no ontology *of substance*"; the disclaimer, where it belongs, localises to the *formal* register ("syntactic descriptions without ontological commitment"), not to the cycle as a whole.

(3) New section: Things and Acts — what inhabits the two registers

To be added to the two-register discussion of the full paper.

Acts and Things. An *Act* is what an operation *constructs*: an object obtained, and witnessed, by performing operations — a term, a witnessed bisimulation, a derivation that leans on nothing posited. A *Thing* is what an axiom *posits*: an object, a totality, or an existence principle introduced by *postulation* rather than earned by construction (Hilbert's implicit definition — a Thing is whatever a consistent axiom makes it, relative to that axiom). The two registers are exactly this contrast: the operational register is the register of Acts (built, choice-free, becoming); the formal register is the register of Things (posited, axiomatic, completed).

The one exception: the medium. Not everything unproven is a Thing. To state any axiom, or to derive any Act, one already reasons — with the connectives, the quantifiers, the rule of inference. This bare logic is not a Thing but the *medium*: it cannot be posited-or-not, for it is presupposed by every posit and every derivation, and it is self-affirming — to deny it one must already use it. Every genuine axiom, by contrast, is *optional* (a system may reject it and still reason) and is therefore a Thing. In the cycle's ledger this puts the Act/Thing line at the empty profile: an Act depends on *no* axiom (`[]`); `propext` and `Quot.sound` are already Things, the cheapest ones; `Classical.choice`, the axiom of infinity, the power set as a completed object, the aleph cardinals and their consequences (e.g. Banach–Tarski) are the expensive ones. The operational register is thus an Act built on the cheapest Things; the formal register adds the expensive ones.

How a Thing is admitted. A Thing is admitted not by construction but by a *consistency check*: one exhibits a model — a structure in which the axiom holds (a theory is consistent iff it has a model) — and thereby a *relative* tag Con(base) $\to$ Con(base + Thing). The tag is never absolute: by Gödel's second theorem no consistent system strong enough for arithmetic proves its own consistency. Hence the operational core, being axiom-free (`[]`), inherits the host's consistency and adds no new burden, while a Thing is the first place a consistency question arises at all.

Consistency is necessary but not sufficient; the cost discriminates. Every Thing that has a model is equally *consistent*; consistency does not tell them apart. What does is the *kind of cost* the Thing carries. Three kinds recur across the cycle:

- **strength** — the axiom of infinity: completed ω proves Con(PA), so it cannot be justified from the finite base at all (Gödel II); a genuine leap in consistency strength;
- **impredicativity** — the power set: to speak of "all subsets of ω" one quantifies over the very totality being formed. Its profile is choice-free (`[propext]`) and does *not* reveal the circularity — the impredicativity is a cost the machine profile fails to capture (finding P-1);
- **non-constructivity** — the axiom of choice: profile `[propext, Classical.choice, Quot.sound]` (the choice floor), an unexhibitable witness (no one displays a well-ordering of $\mathbb{R}$), and the non-measurable / Banach–Tarski pathology.

Thus $\wp(\omega)$ is a cheap, safe Thing and AC an expensive, pathological one, though both are consistent: it is the cost, not consistency, that separates them — and, as P-1 shows, not every cost is visible to the axiom profile.

We track, we do not adopt. A Thing is placed in the formal register *with* its tag and its cost; it is not adopted as an operational truth. By the $O \to T \,/\, T \to O$ asymmetry a Thing does not transit down: the completed power set, for instance, is used formally but has no operational correlate — the operational register carries only its *becoming*, the non-enumerable $\wp$. To work with a totality in VR is therefore not to build it but to posit it, exhibit its model, and pay — in the ledger — for what it costs.

Why VR-Forms is otherwise unaffected

VR-Forms (the operational two-register apparatus over VR-Sets, with the conservativity and transit results) does not depend on the removed layer: its content references VR only through the operational reading and the nullary base $\emptyset$, which is unchanged in v1.1.1. The Lean module (`VRCycle.Forms`) contains no reference to the removed objects (`VRBool`, `impl`, A1, A2), and the build is unaffected. The "propositional floor" of the conservativity theorem (`Forms/Conservativity.lean`) is an internal technical layer of Forms, unrelated to VR's former $\to$ operator.

Prepared with the assistance of Claude (Anthropic, Opus 4.8) in the cycle's Variant A architecture; all mathematical content and positions are due to the author, Vitaly Reznik.